

VibeCodium — Presentation

"The Future of Collaborative Coding" Build at the speed of thought — with agentic AI integration, real-time multi-user editing, sandboxed code execution, and a social developer community. All in the browser.

Introduction

What is VibeCodium?

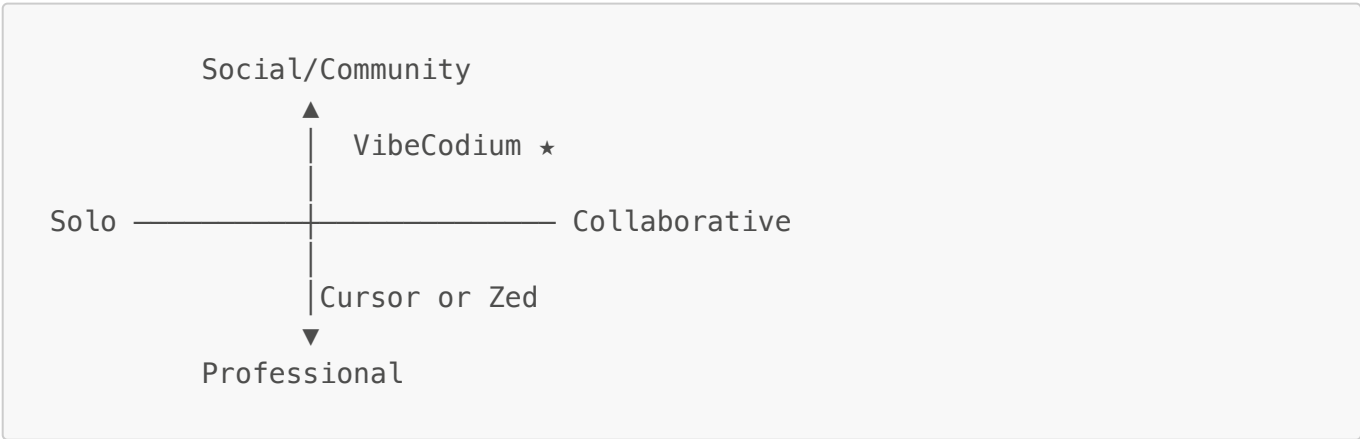
VibeCodium is a **full-stack, cloud-native IDE** — a collaborative code editor that runs entirely in the browser. It combines the power of a professional code editor, an autonomous AI coding agent, real-time multi-user collaboration, and a developer social community into a single, unified platform.

[!IMPORTANT] VibeCodium is **not** just another code editor. It belongs to the 4th generation of development environments — the **AI-Native / Agentic IDE** era, where code is negotiated between the developer and an intelligent agent that reads, writes, and executes in a continuous tool loop.

The Problem We Solve

Problem	Answer
Solo IDEs are isolated — no real-time collaboration	Conflict-free multi-user editing
AI assistants are opaque black boxes	Agent shows its work: Accept/Reject diffs inline
Arbitrary code execution is dangerous	Docker sandbox: network off, 2GB RAM
The lack of community in the professional coding	CoderMatch, Help Posts, VibeMatch for developers

Competitive Positioning



VibeCodium combines **agentic AI capabilities** + **social collaboration** — a space that lacks competition.

The Landing Page

The landing page at is designed to **wow from the first second**. It features:

- **Cinematic hero section** with a warp-speed star animation background, floating data stream effects, and a 3D-tilted terminal preview showing a live "AI Sentient Optimization" block with **Accept Merge** / **Refactor** buttons.
- **Bento grid feature layout** — large cards for the most impressive features, cascading into a secondary 3-column grid.
- A **VS Code-style status bar footer** (always visible) with "VibeCodium v4.0.2 Stable" branding.

Landing Page Sections

1. **Hero** — Main headline, CTA buttons (**Initialize Workspace** / **Import Repository**), and the animated terminal preview mockup
2. **Feature Bento Grid** — Live Collaboration (large), One-Click Deploy (small), AI Agent full-width card
3. **Secondary Features Grid**

If the user is **already authenticated**, the hero button becomes **Dashboard** and the nav profile button shows their GitHub username.

Navigation Bar

Fixed glassmorphism top bar with:

- **VibeCodium** logo (animated terminal icon)
- Community link → **/community**
- Profile/Login button → **/profile** or Auth0 login redirect

The Dashboard

The dashboard at **/dashboard** is the **Command Center** — the hub between your GitHub repositories and your VibeCodium workspaces.

DashboardHeader (user info, nav)	
Sidebar	Your Repositories (headline + stats)
• Files	[🔍 Search repos bar]
• Recent	— Recent Workspaces (3 cards + Import) —
• Deploy	— Deployed Apps —
• Create	— GitHub Repositories (full list) —
• Match	— All Workspaces (saved projects) —

Dashboard Sections

Stats Header — Total Stars ★ across all repos, total Repository count — displayed in glassmorphism stat cards.

Repository Search — Live fuzzy search across repo names and descriptions, with inline results and a Clear button.

Recent Workspaces — Top 3 most-recently-created projects, displayed as large hover-glow cards. An **Import Project** card (dashed border, animated + icon) triggers the Import Modal.

Deployed Apps — Shows all apps previously shipped to Vercel, with a ● Live badge and **Visit App** link.

GitHub Repositories — Complete list of all user repos fetched from the GitHub API, showing language badge (color-coded dot), star count, fork count, and last-update time. Each row has an **Import** button that clones the repo into a VibeCodium workspace.

All Workspaces — A chronological flat list of every saved workspace (SQLite). Click any row to open that project directly in the editor.

DashboardSidebar

The left sidebar contains navigation links for: scrolling to recent workspaces, scrolling to repos, opening CoderMatch, creating a new repo. A badge shows the count of unread VibeMatch messages (polled every 30 seconds).

The Code Editor (Workspace)

Overview

The main IDE workspace at `/?w=<projectId>` is a **professional-grade code editor** running entirely in the browser, with real-time collaboration, a full PTY terminal, and an AI agent sidebar.

3-Column Layout

Left column: Activity Bar , Panel, Explorer, Search, Git, Fun, Help, Spotify, YouTube

Middle column: Editor, Terminal

Right column: The AI Agent

All panels are **resizable** via drag handles (`react-resizable-panels`).

Monaco Editor Features

- **Syntax highlighting** for all major languages
- **Multi-tab** support (open files in tabs, close with x)
- **Tab context menu** — Close, Close Others, Close All
- **Remote cursor rendering** — colored named cursors from other users in real time

- **Pending Edit Blocks (AI Diffs)** — when the AI proposes a change, it highlights the diff in the editor with inline ✓ **Accept** / ✗ **Reject** buttons
- **Power Mode** (Easter egg) — keystroke sparks when typing
- **Embedded Games** — Flappy Bird / Subway Surfer play in a Picture-in-Picture window inside the editor

WorkspaceTopBar

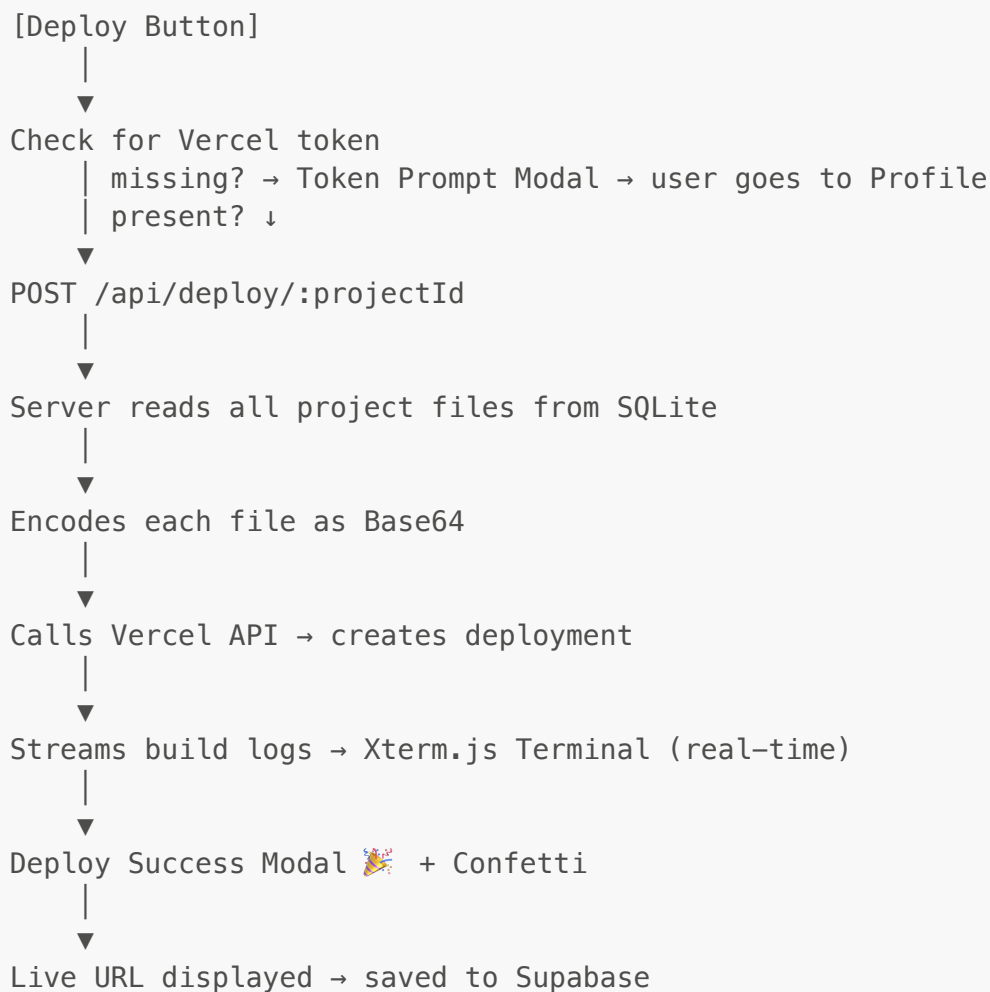
The top bar shows: project name, branch name, collaborator avatars (colored dots), connection status, **Save to GitHub**, **Deploy to Vercel**, **Share Link**, **Whiteboard** toggle buttons.

Ship to Cloud (Vercel Deployment)

One-Click Deploy

VibeCodium integrates the **Vercel API** directly into the editor. With a single click on **Deploy**, the entire project is packaged and shipped to production.

Deployment Flow



Features

- **Real-time terminal streaming** — deployment logs appear in the Xterm terminal as they come in
- **Deploy Success Modal** — shows the live URL with a **Visit App** button and **Copy URL** option
- **Deployed Apps history** — all past deployments appear on the Dashboard's "Deployed Apps" section
- **Token management** — GitHub and Vercel tokens are stored encrypted in Supabase, never in plaintext
- **Confetti celebration** on successful deploy

GitHub Push (Save)

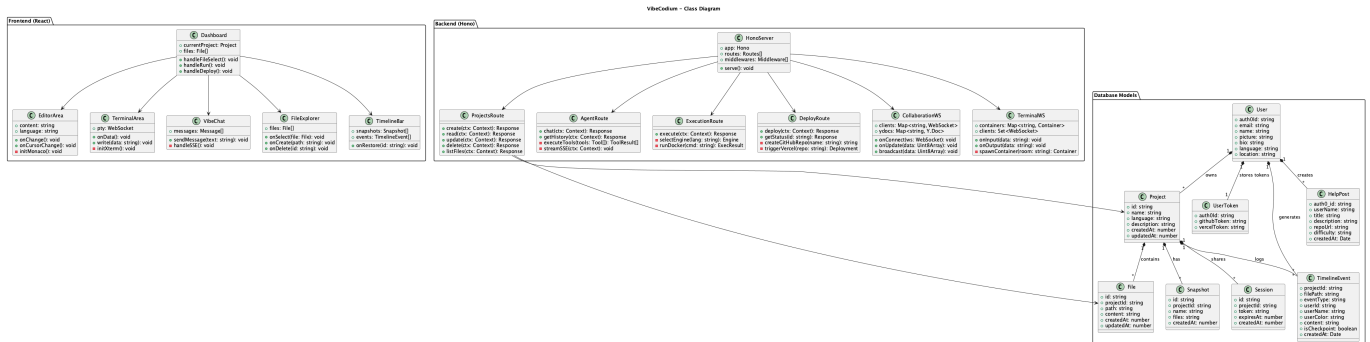
The **Save to GitHub** button in the top bar calls **POST /api/projects/:id/push**, which commits all current file contents back to the linked GitHub repository using the user's stored GitHub token.

[!TIP] Both GitHub token and Vercel token are managed in the **Profile** page under **Integrations**, stored securely.

Real-Time Collaboration

Architecture

VibeCodium's collaboration layer uses **Yjs CRDTs** for conflict-free concurrent editing, transmitted over **WebSockets** via the Bun server.



Client A

User types

```

└─ Y.Doc transact
    └─ doc.on("update")
        └─ WS send {type:"yjs_update"}

```

```

applyUpdate → roomYDocs[proj][file]
update roomFileStates[proj][file]
ws.publish(projectId, {type:"yjs_update"})

```

```

applyUpdate to local Y.Doc
Monaco model.setValue(merged)

```

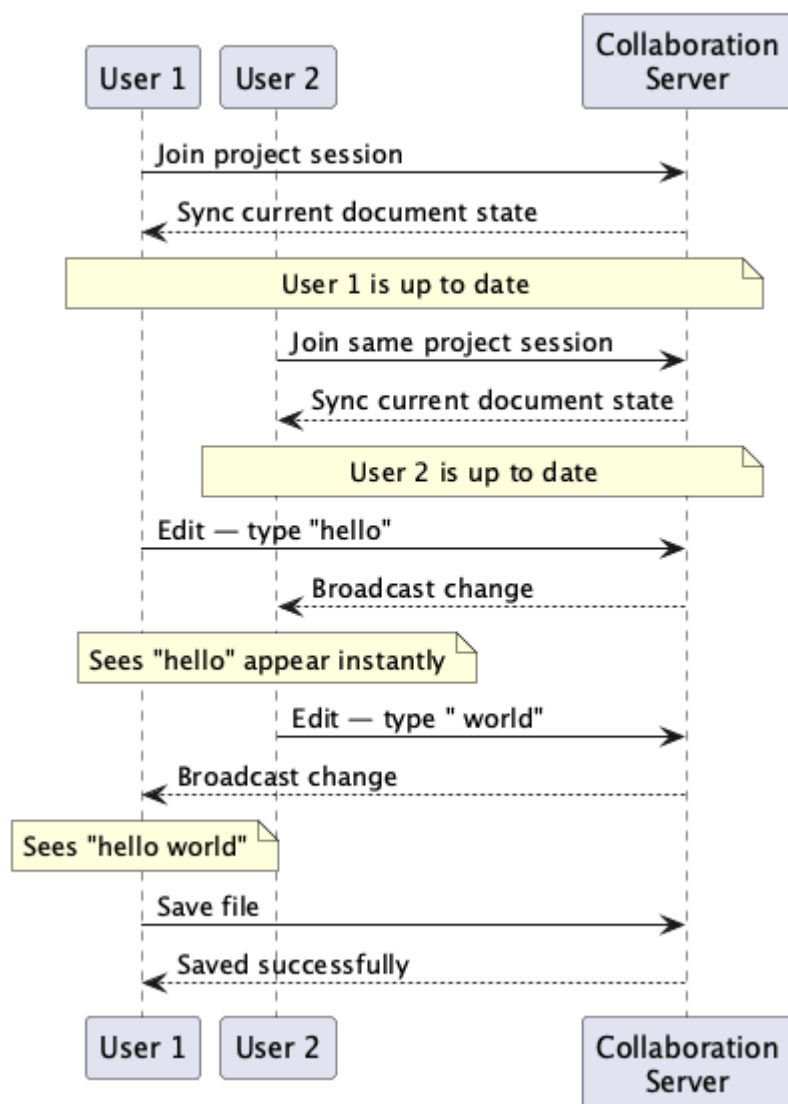
Bun Server (Hono)

Client B

Collaboration Features

- **Colored named cursors** — each collaborator gets a unique color; their cursor and username label appear in the editor in real time
- **Conflict-free merging** — Yjs CRDT guarantees zero conflicts regardless of concurrent edits
- **Room state sync** — when a new user joins, they receive the full current room state immediately
- **User presence list** — the top bar shows avatar dots for every active collaborator
- **Host designation** — the first user to open a project is the "host"; the backend tracks host assignment
- **Agent-accepted sync** — when one user accepts an AI suggestion, the change is broadcast to all collaborators
- **Session sharing** — the **Share Link** button generates a shareable URL (`/?w=projectId`) valid for anyone with the link

VibeCodium – Real-Time Collaboration



Shared Terminal

The Xterm.js terminal is backed by a **real PTY process** (`node-pty`) over WebSocket:

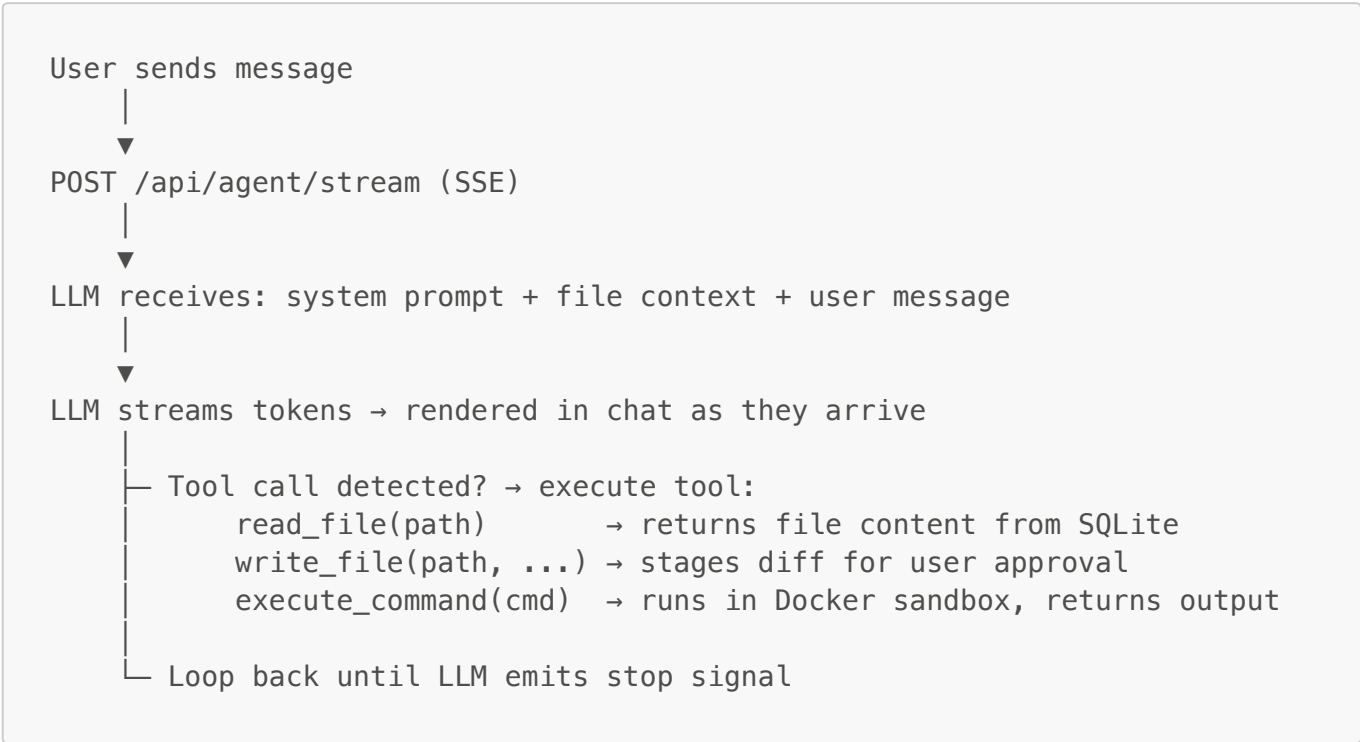
- Every keystroke is sent to the server's PTY
- All output is broadcast to every user in the session
- **Everyone sees the same terminal output simultaneously** — true pair-programming terminal

- Run `npm install`, `python main.py`, `bun run build` — all collaborators see it live

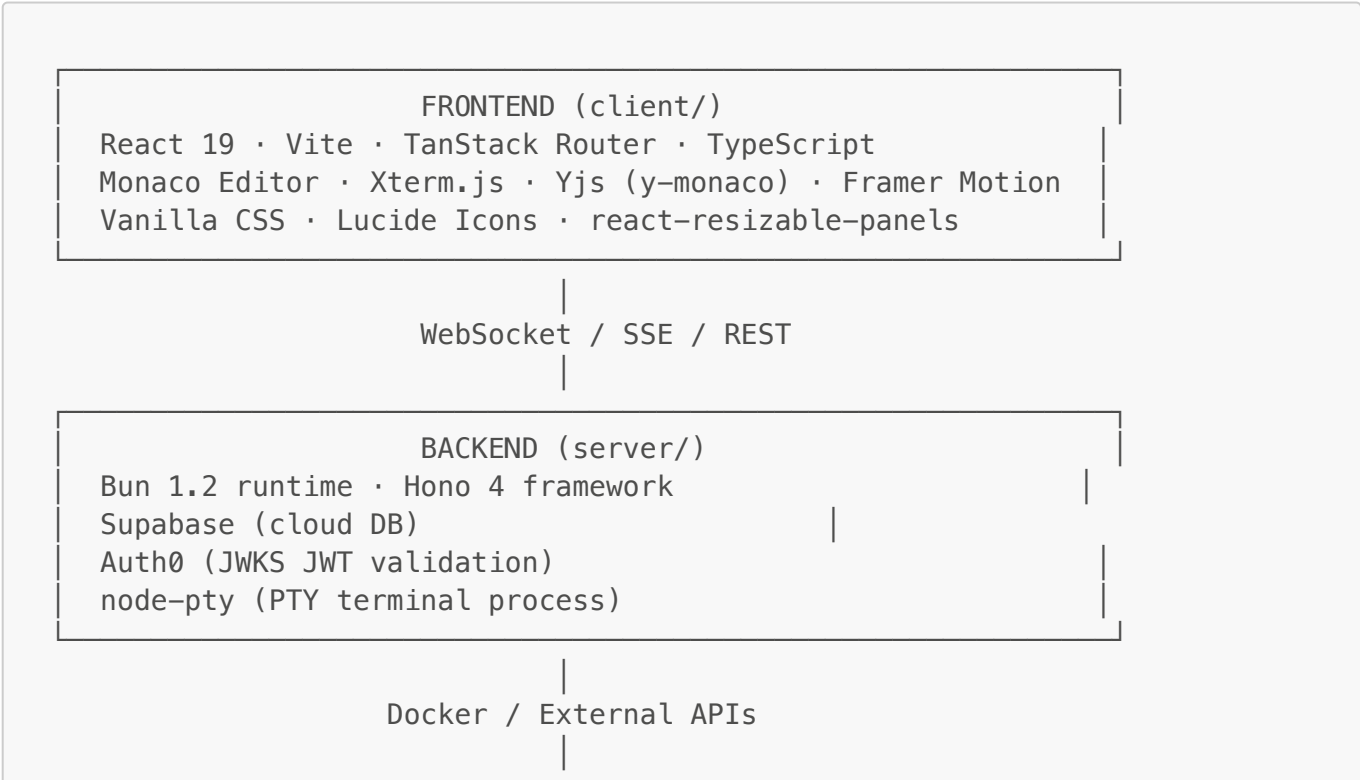
AI Agent (VibeChat)

VibeChat is the AI agent sidebar — a token-streaming, tool-using autonomous coding assistant that lives in the right panel of every workspace.

How the Agent Works



🔧 Technology Stack



Docker Sandbox 6 images Python Node Rust Go C++ Bun	DeepSeek / OpenAI LLM API OpenRouter Groq Custom/ Local	Vercel Deploy API Supabase Atlas (cloud)	GitHub API (repos) YouTube API (Reels)
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Monorepo Structure

```
vibecodium/  
├── client/      React 19 + Vite SPA  
├── server/      Hono API (Bun runtime)  
├── shared/      Shared TypeScript types  
├── turbo.json   Turborepo config  
└── package.json Bun workspaces root
```

VibeMatch, Community & Profile

VibeMatch (CoderMatch)

VibeMatch is a Tinder-style developer matchmaking feature that pairs developers together for collaborative coding sessions.

How It Works

- 1. **Set up your profile** — bio, primary language, location (in the Profile page, under "Public Profile for Vibe Match")
- 2. **Browse & swipe** — the CoderMatch modal shows other developers (profile picture, bio, language, location)
- 3. **Match!** — when two users both swipe right on each other, a match is created
- 4. **Chat** — matched users can message each other directly in the app
- 5. **Unread badge** — the Dashboard sidebar shows a live unread message count, polled every 30 seconds

Community Page (/community)

The community page has two sections:

Help Posts — Developers can post repositories for community code review. Other users can browse, reply, and offer help. Posts link directly to GitHub repos.

CoderMatch Hub — Full VibeMatch interface for browsing and matching with developers.

Vibe Reels (YouTube Sidebar)

A TikTok-style **short-form video feed** of coding content, accessible from the YouTube icon in the editor's activity bar. Videos are cached via the YouTube Data API v3.

Profile Page (/profile)

The profile page is a full settings hub:

Profile Card (Left Column)

- Auth0 avatar image (with glowing online indicator)
- Full name + GitHub username + email
- **Verified Dev** badge
- **Edit Profile** (scrolls to form) and **Disconnect** (logout) buttons

GitHub Stats

Fetches live from GitHub API:

- **Repositories** count
- **Commits** count (total across all repos)

Public Profile (for VibeMatch)

- **Bio** (up to 200 characters)
- **Primary Language** (dropdown: TypeScript, Python, Go, Rust, C++, etc.)
- **Location** (text field: City, Country)
- **Save Profile** button

Integrations

- **GitHub Token** — for pushing to repos and creating repos from the IDE
- **Vercel Token** — for one-click deployment
- **AI Model Provider** — choose from OpenRouter, OpenAI, Groq, DeepSeek, or Custom
 - Set the **Provider**, **Model** (e.g. **anthropic/claude-opus-4**), **API Key**, and **Base URL**

Easter Eggs, Fun Features & Technology Stack



Easter Eggs & Fun Features

VibeCodium is packed with hidden and not-so-hidden fun features, accessible from the **Fun panel** (<⚡> icon in the activity bar).



Power Mode

Toggle ON to enable **keystroke sparks** — every key press generates animated particle effects bursting from the cursor. Great for showing off during a demo.

Hacker Mode (Matrix Rain)

Toggle ON to overlay the entire editor with the classic **Matrix digital rain** effect — falling green katakana characters. Uses a canvas overlay that does not interfere with editing.

Flappy Bird (Picture-in-Picture)

Launch a **fully playable Flappy Bird** game in a floating PIP window that stays on top of the editor. Keep coding while you play.

Subway Surfer (Picture-in-Picture)

Launch a **3D Subway Surfer** game in a PIP window. The classic "coding with brainrot content" experience.

Roast My Code

With any file open, click **Roast My Code** to send the current file to the AI for a **brutally honest (and funny) code review**. The AI generates roast-style commentary in a modal.

Emoji Reactions

Users in a collaboration session can send **floating emoji reactions** that appear on everyone's screen simultaneously (broadcast via WebSocket `emoji_reaction` type).

Rubber Duck (RubberDuck.tsx)

A rubber duck debugging companion component.

Quick Open (Ctrl+P**)**

VS Code-style command palette — type to fuzzy-find any file by name across the entire project.

Zen Mode (Ctrl+K**)**

One shortcut to hide the sidebar, terminal, and chat — leaving only the pure editor in full focus mode.

Spotify Integration

A Spotify player embedded directly in the editor sidebar. Code and vibe simultaneously.

Vibe Reels

TikTok-style coding short videos inside the IDE — because why leave to watch YouTube?